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Let's learn about **satellites**!

Space mission and **satellite** design course

Satellite Subsystems

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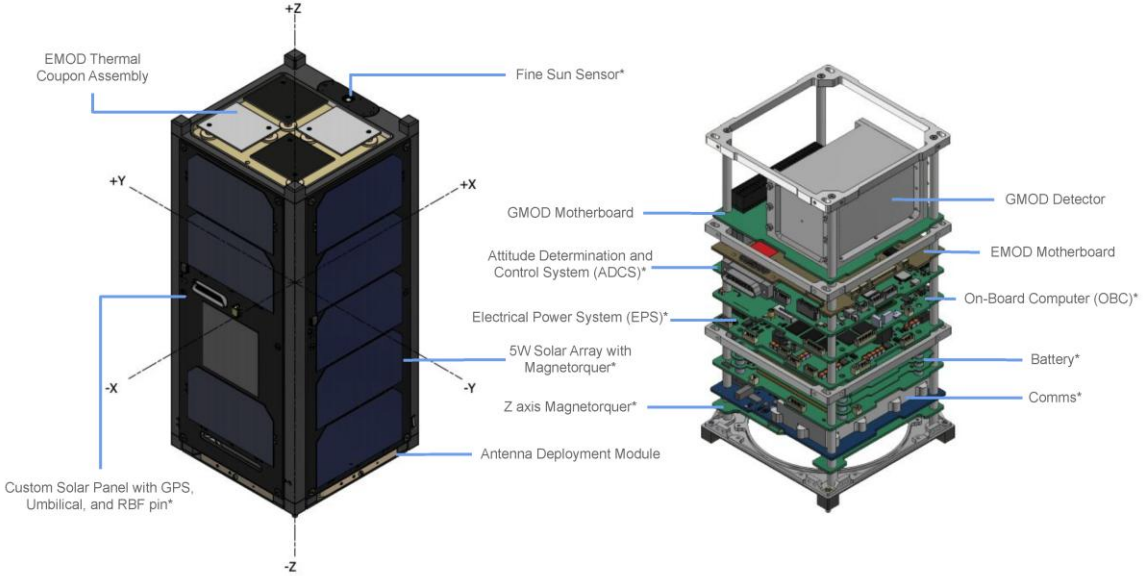
A System of Systems

To perform operations in space with limited or no human intervention, a satellite is composed of multiple subsystems, each with a specific function.

A satellite is a system of systems.



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Satellite Subsystems

A satellite is not a single device.

It is the integration of many subsystems working together.

Like a smartphone, but operating in space.

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Main Satellite Subsystems

Typical satellite **subsystems** include:

- Structure
- Power System
- Communication System
- On-Board Computer (OBC)
- Sensors
- Attitude Determination and Control System (ADCS)
- Navigation System
- Propulsion System
- Thermal Control System
- Payload
- On-board Software
- Interfaces (electrical and mechanical)

Each one is essential!

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Satellite Design

Satellite design involves many disciplines:

- Mechanical Engineering
- Electronic Engineering
- Software Engineering
- Telecommunication Engineering
- Materials Engineering
- Mathematics and Physics
- Aerospace Engineering

You do not need to be an aerospace engineer to work in the space industry!

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Structure

The structure is the mechanical backbone of the satellite.

It must:
- Contain all subsystems
- Withstand launch loads
- Survive vibration and shock
- Distribute thermal loads
- Handle electrostatic discharge

Without structure, nothing holds together.

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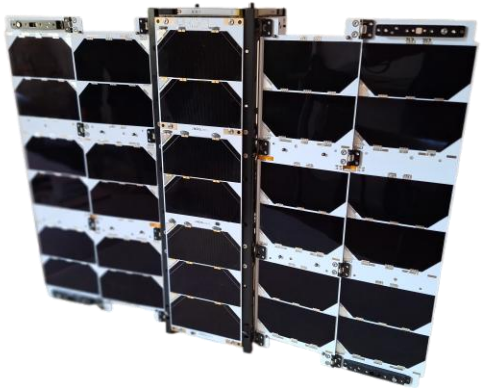
Power System

The Power System provides electrical energy to the satellite.

It includes:
- Power generation (**solar panels**)
- Energy storage (**batteries**)
- **Power management and distribution**

It must ensure:
- Correct voltage levels
- Safe battery charging
- Protection against failures

No power = no mission.



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Communication System

The Communication System connects the satellite to Earth.

It allows:
- Telemetry downlink
- Telecommand uplink
- Payload data transmission

Typically composed of:
- **Transceiver** (TX/RX module)
- **Antennas**

Without communication, the satellite becomes space debris.

Some satellites use more than one radio system!

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On-Board Computer (OBC)

The **OBC** is the brain of the satellite.

It:
- Controls all subsystems
- Executes commands
- Collects telemetry
- Manages payload operations

It contains:
- Microprocessor
- Volatile memory
- Non-volatile memory
- Interface controllers

<u>All decisions pass through the OBC.</u>

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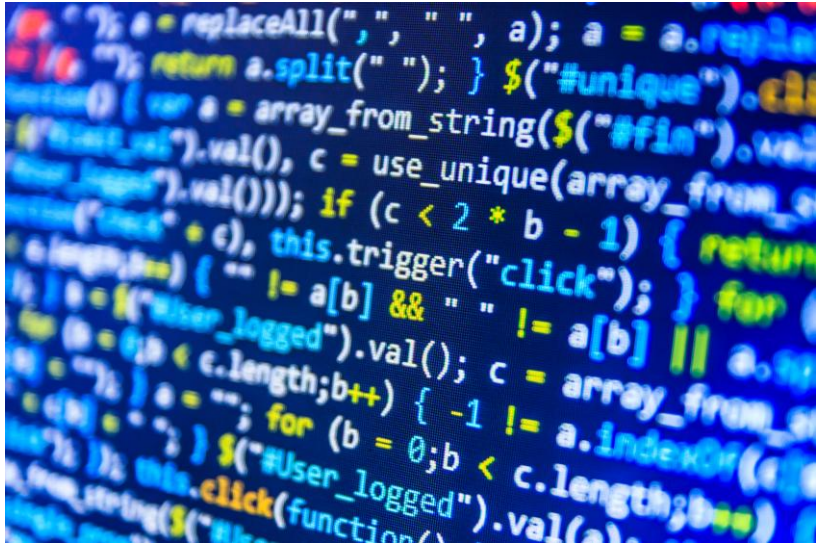
On-Board Software

On-board software runs inside the OBC.

It performs:
- Attitude control algorithms
- Telecommand execution
- Telemetry collection
- Time synchronization
- Failure detection and recovery

Software defines the **satellite behavior**.



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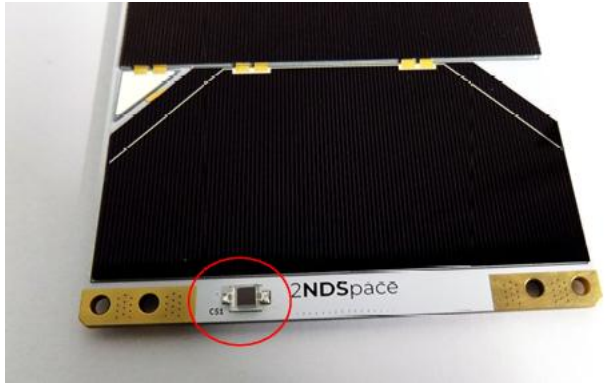
Sensors

Sensors determine the satellite attitude and state.

Examples:
- Gyroscopes
- Accelerometers
- Magnetometers
- Star Trackers
- Sun sensors
- Thermal sensors

Many sensors must be placed externally to interact with the space environment.



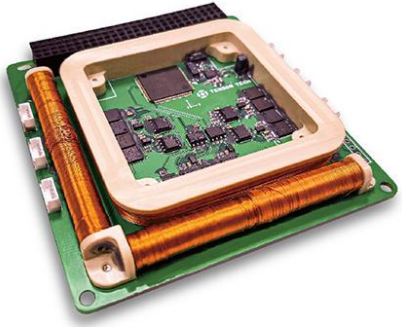
Attitude Control System (ADCS)

ADCS controls the orientation of the satellite.

It includes:
Attitude Determination (sensors)
Attitude Control (actuators)

Typical actuators:
- Magnetorquers
- Reaction wheels
- Cold gas thrusters

Important:
ADCS changes orientation, not the orbit.



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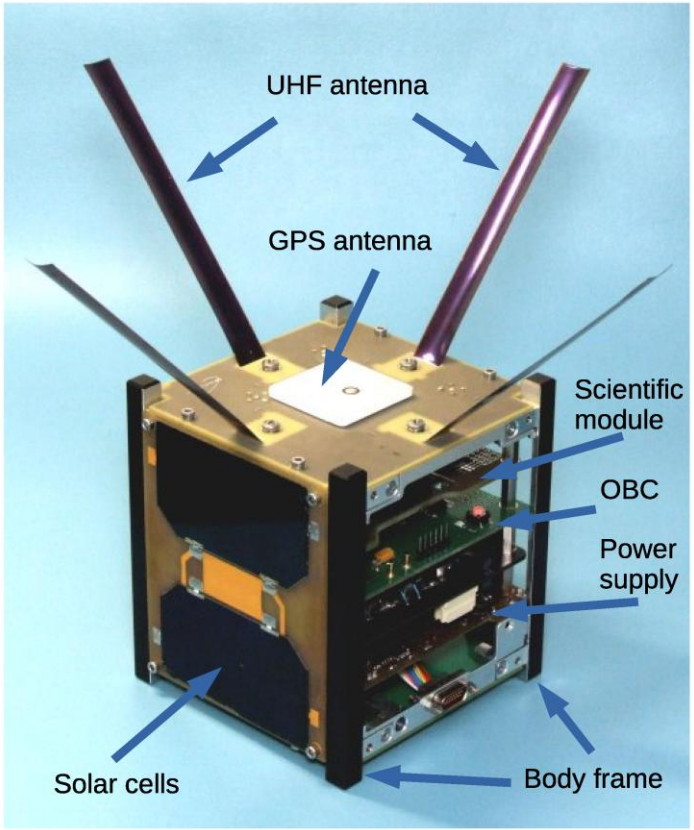
Navigation System

Navigation determines the satellite position in orbit.

Usually based on:
- GPS
- Galileo
- GLONASS

It provides orbital position and velocity data.



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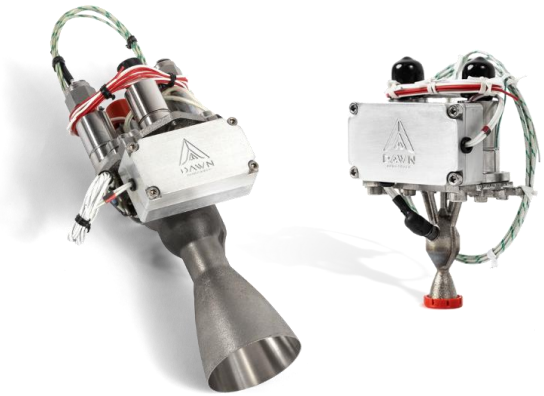
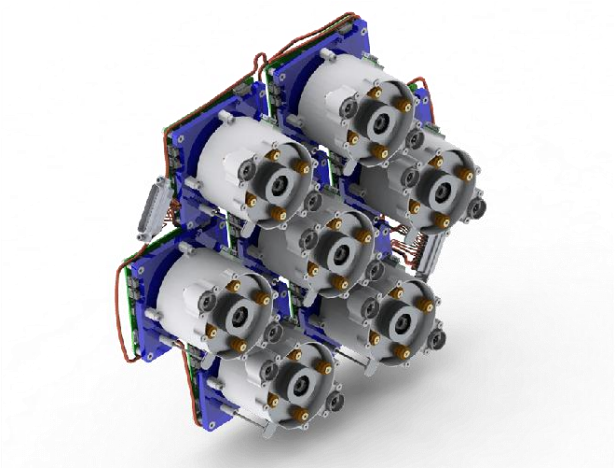
Propulsion System



Propulsion allows the satellite to change orbit.

It can be used for:
- Orbital transfer
- Station keeping
- De-orbiting

Unlike ADCS, propulsion changes the orbit.



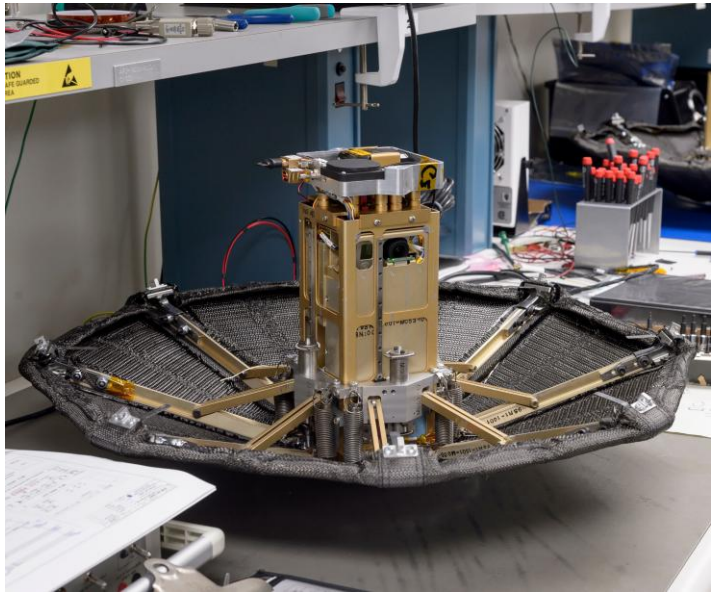
Thermal Control System

Space is a **harsh thermal environment**.

The Thermal Control System:
- Maintains temperature limits
- Prevents overheating
- Avoids excessive cooling

Two types:
Passive systems
- Radiators
- Heat paths
- Surface coatings

Active systems
- Pumps
- Fluid loops
- Heaters



Electronics must operate within strict temperature ranges.

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Internal vs External Subsystems

Some subsystems are **internal**:
- OBC
- Batteries
- Electronics

Others are **external**:
- Solar panels
- Antennas
- Sensors

External hardware must survive **radiation**, **vacuum**, and **extreme temperatures**.

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Mechanical Integration (PC104)

Internal electronics can be integrated using standards.

In CubeSats, a common standard is: **PC104**

It defines:
- Mechanical stacking
- Electrical connectors
- PCB dimensions

Standardization simplifies integration and it's the main advantage of CubeSats.

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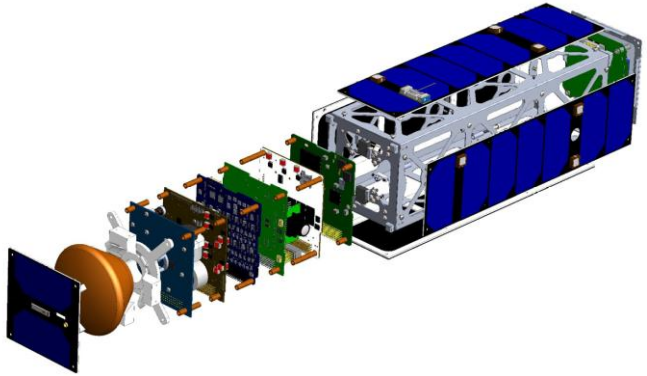
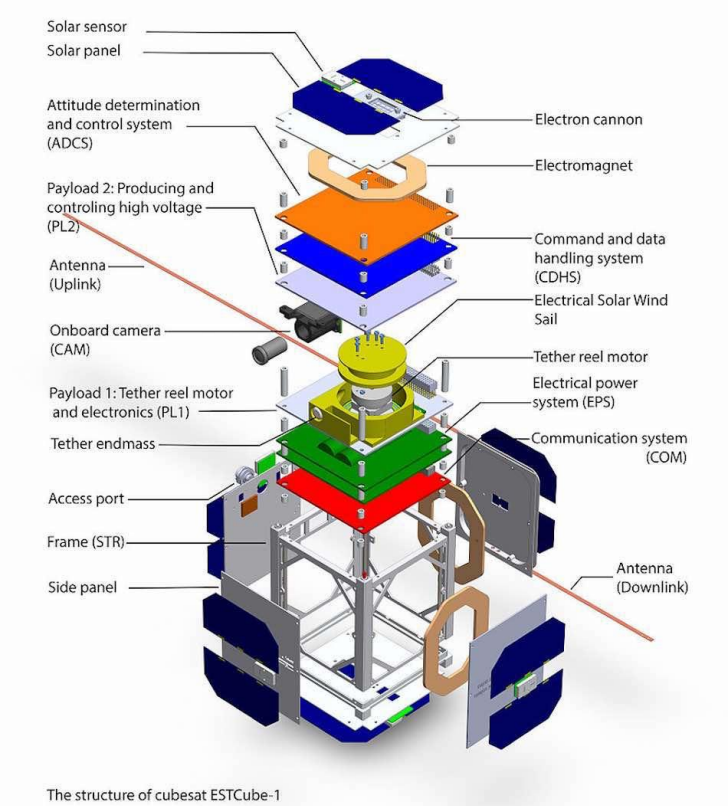
CubeSat Example

A CubeSat integrates all subsystems within a very small volume.
Everything must fit inside strict mass, volume and power budgets.
Miniaturization drives innovation.





The structure of cubesat ESTCube-1

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Satellite Design Software

Satellite development requires several **software tools**.

Categories:
- Mission analysis software
- Engineering design software
- Simulation tools
- Analysis software
- Project management software

Modern satellites are designed digitally.

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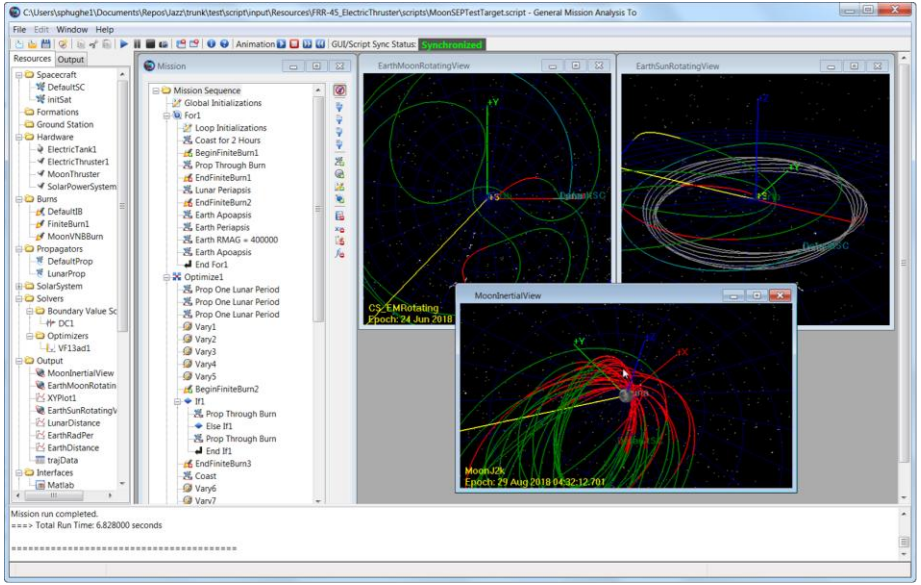
Mission Analysis

Mission analysis tools allow:
- Orbit design
- Coverage analysis
- Ground station visibility
- Lifetime estimation

Examples:
- STK
- GMAT

Before building hardware, we simulate the mission.



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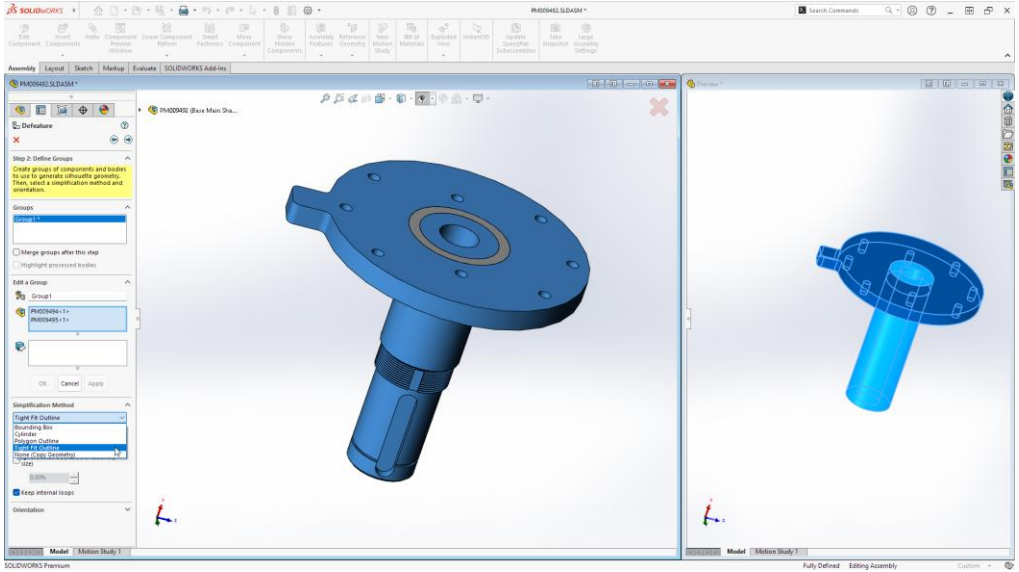
Mechanical Design

Mechanical **CAD** tools allow:
- 3D modeling
- Assembly checks
- Interference detection
- Mass and inertia analysis

Examples:
- CATIA
- SolidWorks
- PTC Creo
- Fusion 360



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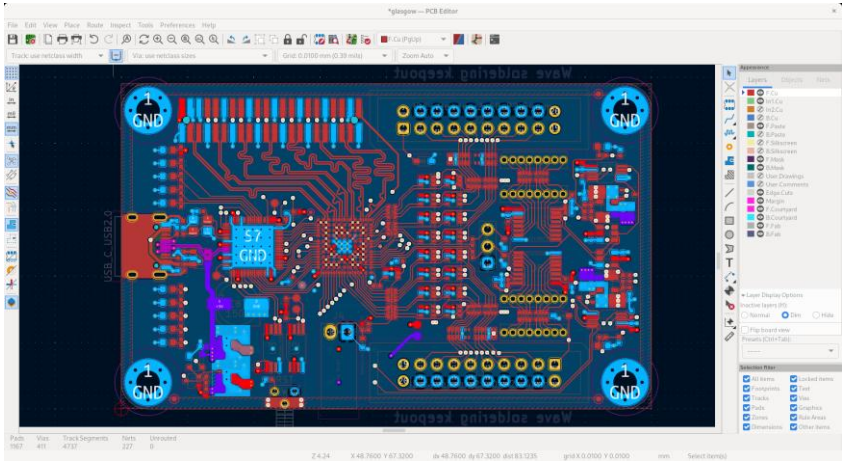
Electrical Design

Electrical design tools allow:
- Schematic design
- PCB layout
- Production file generation
- 3D PCB export

Examples:
- KiCad
- Altium
- Eagle



Electronics must integrate mechanically and electrically.

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Concurrent Engineering

Subsystem teams work in parallel.

Mechanical, electrical and software teams develop simultaneously.

This reduces development time. But requires strong coordination.

Space missions are **built step by step**.

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